

# Homework 6 Solutions

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## Problem 14.1

- a)  $\{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$
- b)  $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 4), (3, 3), (4, 4), (5, 5)\}$
- c)  $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$
- d)  $\{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5), (2, 2), (2, 4), (4, 2), (4, 4)\}$

## Problem 14.2

- a)  $y$  is one greater than  $x$ .
- b)  $x \geq y$ .
- c)  $x + y$  is equal to 6.
- d)  $y$  is divisible by  $x$ .

## Problem 14.3

- a) reflexive, symmetric, antisymmetric, transitive
- b) irreflexive, antisymmetric
- c) antisymmetric, transitive
- d) symmetric
- e) reflexive, symmetric, transitive

## Problem 14.4

- a) reflexive, symmetric, transitive
- b) irreflexive, antisymmetric
- c) reflexive, symmetric, transitive
- d) reflexive, symmetric
- e) irreflexive, symmetric
- f) irreflexive, antisymmetric, transitive

## Problem 14.5

*Proof.* The  $=$  relation on the integers is antisymmetric.

1. Let  $x, y \in \mathbb{Z}$  such that  $x = y$  and  $y = x$ .
2. Then  $x = y$ .
3. Therefore the  $=$  relation is antisymmetric.

□

### Problem 14.6

a)  $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : |x - y| \leq 2\}$

b) *Proof.*  $R$  is reflexive.

1. Let  $(x, y) \in \mathbb{Z} \times \mathbb{Z}$  such that  $y = x$ .
2. Then  $|x - y| = |x - x| = 0$ , so  $|x - x| \leq 2$ .
3. Therefore  $(x, x) \in R$ .
4. Therefore  $R$  is reflexive. □

c) *Proof.*  $R$  is not irreflexive.

1. Let  $(x, y) = (1, 1)$ . 2. Observe that  $|1 - 1| \leq 2$ .
3. Therefore  $(1, 1) \in R$ .
4. Therefore  $R$  is not irreflexive. □

d) *Proof.*  $R$  is symmetric.

1. Let  $(x, y) \in R$ .
2. Then  $|x - y| \leq 2$ .
3. Then  $|x - y| = |-1||x - y| = |y - x|$ , so  $|y - x| \leq 2$ .
3. Therefore  $(y, x) \in R$ .
4. Therefore  $R$  is symmetric. □

e) *Proof.*  $R$  is not antisymmetric.

1. Let  $(x, y) = (1, 2)$ .
2. Observe that  $|1 - 2| \leq 2$ , and  $|2 - 1| \leq 2$ .
3. Therefore  $(1, 2)$  and  $(2, 1) \in R$ , but  $1 \neq 2$ .
4. Therefore  $R$  is not antisymmetric. □

f) *Proof.*  $R$  is not transitive.

1. Let  $(x, y) = (1, 3)$ , and  $(y, z) = (3, 5)$ .
2. Observe that  $|1 - 3| \leq 2$ , and  $|3 - 5| \leq 2$ , so  $(x, y)$  and  $(y, z) \in R$ .
3. Observe that  $|1 - 5| > 2$ , so  $(x, z) \notin R$ .
4. Therefore  $R$  is not transitive. □

### Problem 14.7

a)  $R^{-1} = \{(2, 1), (3, 2), (4, 3)\}$

b)  $R^{-1} = \{(1, 1), (2, 2), (3, 3)\}$

c)  $R^{-1} = \{(x, y) : x, y \in \mathbb{Z}, y - x = 1\}$

d)  $R^{-1} = \{(x, y) : x, y \in \mathbb{N}, y|x\}$

e)  $R^{-1} = \{(x, y) : x, y \in \mathbb{N}, xy > 0\}$

### Problem 14.8

*Proof.* If  $R$  and  $S$  are relations such that  $R = S^{-1}$ , then  $S = R^{-1}$ .

1. Let  $R$  and  $S$  be relations such that  $R = S^{-1}$ .
2. Let  $(x, y) \in S$ .
3. Then  $(y, x) \in S^{-1}$ , and since  $R = S^{-1}$ , then  $(y, x) \in R$  and so  $(x, y) \in R^{-1}$ .
4. Therefore  $S \subseteq R^{-1}$ .
5. Now, let  $(x, y) \in R^{-1}$ .
6. Then  $(y, x) \in R$ , and since  $R = S^{-1}$ , then  $(y, x) \in S^{-1}$  and so  $(x, y) \in S$ .
7. Therefore  $R^{-1} \subseteq S$ .
8. Therefore  $S = R^{-1}$ . □

### Problem 14.15

*Proof.* A relation  $R$  on a set  $A$  is antisymmetric iff  $R \cap R^{-1} \subseteq \{(a, a) : a \in A\}$ .

1. ( $\Rightarrow$ ) Let  $R$  be an antisymmetric relation on  $A$ , and let  $E = \{(a, a) : a \in A\}$ .
2. Let  $(x, y) \in R \cap R^{-1}$ .
3. Then  $(x, y) \in R$  and  $(x, y) \in R^{-1}$ .
4. Since  $(x, y) \in R^{-1}$  then  $(y, x) \in R$ .
5. Since  $(x, y) \in R$  and  $(y, x) \in R$ , and since  $R$  is antisymmetric, then  $x = y$ .
6. Then  $(x, y) = (x, x)$ , and since  $R$  was a relation on  $A$ , then  $(x, y) \in E$ .
7. Therefore  $R \cap R^{-1} \subseteq E$ .
8. ( $\Leftarrow$ ) Let  $E = \{(a, a) : a \in A\}$ , and let  $R$  be a relation on  $A$  such that  $R \cap R^{-1} \subseteq E$ .
9. Let  $(x, y) \in R$  such that  $(y, x)$  is also in  $R$ .
10. Then  $(x, y) \in R^{-1}$ .
11. Since  $(x, y) \in R$  and  $(x, y) \in R^{-1}$ , then  $(x, y) \in R \cap R^{-1}$ .
12. Since  $R \cap R^{-1} \subseteq E$ , then  $(x, y) \in E$ .
13. Therefore  $x = y$ .
14. Therefore  $R$  is antisymmetric.
15. Therefore a relation  $R$  on a set  $A$  is antisymmetric iff  $R \cap R^{-1} \subseteq \{(a, a) : a \in A\}$ . □